

Problem Set 2

Instructor: Bill Fefferman

1 More set theory

Problem 1: Recall that a function $f : A \rightarrow B$ is injective if $\forall x, x' \in A, f(x) = f(x') \Rightarrow x = x'$ and we say a function $f : A \rightarrow B$ is surjective if $\forall y \in B \exists x \in A$ so that $y = f(x)$. Using functions from the integers to the integers (i.e., $f : \mathbb{Z} \rightarrow \mathbb{Z}$), give an example of:

- A function that is injective but not surjective.
- A function that is surjective but not injective.

Problem 2: We say a function $g : B \rightarrow A$ is the inverse of a function $f : A \rightarrow B$ if:

1. $\forall x \in A, g(f(x)) = x$ and
2. $\forall y \in B, f(g(y)) = y$.

Notice that not all functions have a well-defined inverse function! In fact, prove that a function has an inverse if and only if it is a bijection (recall, a bijection is a function that is both injective and surjective).

Problem 3: Recall we denote “set difference” by the notation $A \setminus B$. That is, all elements of the set A not in set B . Use a Venn diagram to “prove” that set difference is not associative. That is, show that $(A \setminus B) \setminus C \neq A \setminus (B \setminus C)$ (proving this without the picture is a bit involved, so I won’t require it here). Use the intuition gained from the Venn diagram to give a necessary and sufficient condition involving the sets A, B, C under which set difference is associative.

2 Number theory

Problem 4: Use the Euclidean algorithm to find $GCD(84, 63)$. Write out each step explicitly.

Problem 5: Given positive integers x, y, z does $x \mid (y \cdot z) \Rightarrow x \mid y$ or $x \mid z$? Give an example/counterexample.

Problem 6: Prove/Disprove the following claim: For all integers a, b if $3a - 5b = 31$ then $\gcd(a, b) \neq 17$.